

Element I:

Prototype Testing Data

Sturdy/sufficient:

The test for this criteria will be to see if the box can withstand pressure from both the inside and outside. One method to measure the durability will be to have a various group of people strike the box. Each person will demonstrate different forces which will replicate different forces from different komodo dragons.

	Subject 1	Subject 2
Did it hold?	yes	yes

Feedback: The subjects were worried that the sides would not hold because of how flimsy they were prior to being screwed in. However, once they were finally screwed in, the subjects were pleasantly surprised at how well it held. They stated that it would be able to withstand a large strike.

Comfortable handles:

A basic exam for this criteria is to host a group of people who will rate whether the handles are comfortable. The rating will range from 1 to 5. The rating of 1 will be that they do not want to

hold the handle for more than a few seconds, and 5 being that they would willingly hold it for a long amount of time.

	Subject 1	Subject 2
Rating	4	3

Feedback: Both subjects explained how the circular shape of the handle was good for ergonomics. However, there are multiple paint chips and drips which cause uneven textures.

Ability/inability to open from the inside:

The gate should not be able to open from the inside as it could potentially cause harm to the komodo keepers. The keepers do not want the animal to escape or exit before they are prepared. The ideal test for this would be to have various people with different strengths go into the box and test if it can open. People on the inside of the box won't be able to open the doors. This is because the doors are heavy and they can sometimes get stuck and require a lot of force to pull out again. This can be seen as both a downside because it requires more strength to remove and an upside because the dragons won't be able to get out.

Ability for 2 people to carry:

The Komodo dragons are very heavy, so zookeepers need a light transportation method. If only 2 people are available to be doing the transporting, then the box should be light enough for them to accomplish it. The test for this criteria is to have multiple lineups of 2 people to attempt to carry the box. It is important to have different people of different strengths because the zookeepers will not all be of the same abilities. This will test if they can lift the box

	Group 1	Group 2
yes/no	no	yes

Feedback: The original idea was to have it be light enough for 2 people to carry, however the material used was changed to steel which is heavier. The ability or inability to carry the box depends largely on how strong the group is. The box can be lifted with 2 to 4 people. This was not the ideal ending situation, but with the budget changes, it was the lightest product we could afford.

Easy to clean:

The zookeepers asked that the box be a material which can easily be wiped off. The komodo dragons do not have designated bathroom areas which they use, so it is always possible that they will use the bathroom inside of the box. They can also drag dirt and other mess into the box

which would also cause it to be cleaned. It is important to keep the box clean, not only for look and smell, but also for the welfare of the komodo dragon when using the box. The test will be to have different people wipe up the mess that is made and then rate how easy they thought it was. The scale will range from 1 to 5, meaning that 1 is it was awful to clean and every other surface would be better and 5 being it was very easy to clean.

	Ellison	Bennett
Ease (1-5)	4	4

Feedback: Because of the location of the plastic in the box, it might be harder to ensure that it is clean. However, it was easy to clean because it was a mostly smooth material. This test was concluded after WD-40 was used to drill holes in the box. Because of the steel excess from drilling, the test included both liquid and solid particles, which is essential in a proper test for cleaning. All that was used to clean the boxes was paper towels and a small sweeper to obtain the little particles of metal that came off while drilling the holes.

Folding handles:

As decided in the group's decision matrix, folding handles was the better option when compared to stationary handles. The test will determine how well the handles move. The subjects tested will be rating the ease of handling the handles. The scale will range from 1 to 5. If the handles

are rated a 1, then the handles were squeaky and hard to move, and 5 being that they were very smooth.

Name	Subject 1	Subject 2
Ease (1-5)	4.2	3.4

Feedback: Subject 1 for this test claimed that the handles are uneven and one side of a handle was not sturdy. The handles will need to be adjusted to create a more secure attachment.

Another test for the handles will be to see how much weight they can hold. This will also test how much weight the box can hold.

Can the handles and box support each level of weight?

25lbs: YES 50lbs: YES 100lbs: YES 200lbs: YES 350 lbs: YES

Feedback: The handles were able to hold up to 350 lbs which is all we tested up to. This was done with just one of the handles, so when the box is all put together, having multiple handles ensures that lots of weight can be held. Colton and Cole attached the handles to one of the robotics carts in the multipurpose room that Colt knows weighs over 300 lbs and the handle didn't shake or sway at all. The cart lifted off of the ground on the side with the handle.



This shows the handle attached by two hinges to a robotics cart full of metal and plastics. The handle was comfortably able to lift the side off of the ground.